

Experience clearly indicates that it is generally easier and better to convert an expert in a given field into a partial expert in computing than it is to try to make a computing expert into an expert in the given field. But if we are to require this, then it falls on the computing experts to make every effort to reduce the burdens of learning and using the computer. Arbitrary rules, special jargon, meaningless forms, changes in the methods and form, and delays in access to the machine should all be reduced to a minimum and carefully monitored to reduce them further when the next machine offers new opportunities to lift from the outsider the burden of the nonessentials of computing.

The study of numerical methods and library routines to be used to increase insight, rather than mere machine efficiency, is still in its infancy, and it is one of the most important areas of future research. To work in this field requires experience in the use of computing in everyday work. It is, however, a field well worth cultivating. Clearly the author does not subscribe to the "black box" theory of library routines. The author does, however, subscribe to the heresy that *some* of the results found in the field of artificial intelligence¹ could profitably be applied to *some* numerical analysis routines.

N+1.10 CLOSING REMARKS

It should not be necessary to remind the reader that most of the above remarks are personal opinions developed by the author while working in a particular laboratory, and they do not necessarily have universal applicability. But in defense of them, they seem to be grounded in common sense as well as in experience. If the reader does not like them, he should not argue the topic but should try to develop his own. It is important for the progress of machine computation that the intuitive methods that we now use become more clearly understood and reduced, when possible, to explicit formulations suitable for use with a computer.

¹ See, for example, Nilsson [45] and Slagle [52].

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